Microservice Architecture (/index.html)

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Pattern: Client-side UI composition

pattern (/tags/pattern)

Context

You have applied the Microservice architecture pattern (../microservices.html). Services are developed by business capability (../decomposition/decompose-by-business-capability.html)/subdomain (../decomposition/decompose-by-subdomain.html)-oriented teams that are also responsible for the user experience. Some UI screens/pages display data from multiple service. Consider, for example, an Amazon-style product detail page, which displays numerous data items including:

- Basic information about the book such as title, author, price, etc.
- Your purchase history for the book
- Availability
- Buying options
- Other items that are frequently bought with this book
- Other items bought by customers who bought this book
- Customer reviews
- Sellers ranking
- …

Each data item corresponds to a separate service and how it is displayed is, therefore, the responsibility of a different team.

Problem §

How to implement a UI screen or page that displays data from multiple services?

Forces

Solution

Each team develops a client-side UI component, such an AngularJS directive, that implements the region of the page/screen for their service. A UI team is responsible implementing the page skeletons that build pages/screens by composing multiple, service-specific UI components.

Related patterns

- The Server-side page fragment composition (server-side-page-fragment-composition.html) pattern is an alternative approach

pattern (/tags/pattern)

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